

Zepeng Zhang

Intelligent Maintenance and Operations Systems (IMOS) Lab
École Polytechnique Fédérale de Lausanne (EPFL)

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[zepengzhang.github.io](https://github.com/zepengzhang)

EDUCATION

- Ph.D. Electrical Engineering, École Polytechnique Fédérale de Lausanne, 2023.09-present
Supervisor: Prof. Olga Fink
- M.S. Information and Communication Engineering, ShanghaiTech University, 2020.09-2023.07
Overall GPA: 3.95/4.00; Major GPA: 4.00/4.00;
- B.Eng. Measuring & Control Technology and Instrumentations, Wuhan University, 2016.09-2020.05

EXPERIENCES

- 2025.04–2025.08 Visiting Student, [Télécom Paris, Institut Polytechnique de Paris](#)
- 2023.02–2023.04 Research Intern, [Peking University](#)
- 2019.12–2020.09 Visiting Student, [ShanghaiTech University](#)
- 2019.07–2019.09 Visiting Student, [City University of HongKong](#)
- 2018.01–2020.06 Member, [WHUAI \(ZhuYing Team\)](#)

CURRENT RESEARCH INTERESTS

Graph Machine Learning, Graph Signal Processing, Generative Model

PUBLICATIONS

1. Spatiotemporal Imputation with Graph-Informed Flow Matching
Zepeng Zhang, Aref Einizade, Jhony H. Giraldo and Olga Fink
International Conference on Machine Learning (ICML) 2026
2. MM-Net: Recurrent Unfolding with Adaptive Majorization for Weighted Sum-Rate Beamforming
Zhexian Yang, **Zepeng Zhang** and Ziping Zhao
IEEE Signal Processing Letters 2026
3. [From Physics to Machine Learning and Back: Part II - Learning and Observational Bias in Prognostics and Health Management \(PHM\)](#)
Olga Fink, Ismail Nejjar, Vinay Sharma, Keivan Faghieh Niresi, Han Sun, Hao Dong, Chenghao Xu, Amaury Wei, Arthur Bizzi, Raffael Theiler, Yuan Tian, Leandro Von Krannichfeldt, Zhan Ma, Sergei Garmaev, **Zepeng Zhang**, Mengjie Zhao, Kevin Steiner, Yusuf Kesmen
Reliability Engineering & System Safety (RESS) 2026

4. **From Physics to Machine Learning and Back: Part I - Learning with Inductive Bias in Prognostics and Health Management (PHM)**
Olga Fink, Vinay Sharma, Ismail Nejjar, Leandro Von Krannichfeldt, Sergei Garmayev, **Zepeng Zhang**, Amaury Wei, Gaetan Frusque, Florent Forest, Mengjie Zhao, Chi-Ching Hsu, Keivan Faghhi Niresi, Han Sun, Hao Dong, Chenghao Xu, Raffael Theiler, Arthur Bizzi, Kevin Steiner
Reliability Engineering & System Safety (RESS) 2026
5. Weighted Sum Secrecy Rate Maximization via Minorization-Maximization
Zhexian Yang, **Zepeng Zhang** and Ziping Zhao
IEEE International Symposium on Information Theory (ISIT) 2026
6. **Domain Adaptive Unfolded Graph Neural Networks**
Zepeng Zhang and Olga Fink
AAAI Conference on Artificial Intelligence (AAAI) 2025
7. **Algorithm-Informed Graph Neural Networks for Leakage Detection and Localization in Water Distribution Networks**
Zepeng Zhang and Olga Fink
Reliability Engineering & System Safety (RESS) 2025
8. RINS-T: Robust Implicit Neural Solvers for Time Series Linear Inverse Problems
Keivan Faghhi Niresi, **Zepeng Zhang** and Olga Fink
IEEE Transactions on Instrumentation and Measurement (TIM) 2025
9. **GeoLink: Empowering Remote Sensing Foundation Model with OpenStreetMap Data**
Lubin Bai, Xiuyuan Zhang, Siqi Zhang, **Zepeng Zhang**, Haoyu Wang, Wei Qin, and Shihong Du
Conference on Neural Information Processing Systems (NeurIPS) 2025
10. Weighted Sum-Rate Maximization using MM: Convergence Rate and Deep Adaptation
Zhexian Yang, **Zepeng Zhang**, and Ziping Zhao
European Signal Processing Conference (EUSIPCO) 2025
11. **Graph Neural Networks With Adaptive Structures**
Zepeng Zhang, Songtao Lu, Zengfeng Huang, and Ziping Zhao
IEEE Journal of Selected Topics in Signal Processing: Special Series on AI in Signal & Data Science (JSTSP) 2024
12. **Accelerating Quadratic Transform and WMMSE**
Kaiming Shen, Ziping Zhao, Yannan Chen, **Zepeng Zhang**, and Hei Victor Cheng
IEEE Journal on Selected Areas in Communications: Special Issue on Advanced Optimization Theory and Algorithms for Next Generation Wireless Communication Networks (JSAC) 2024
13. Graph Neural Networks With Adaptive Structures
Zepeng Zhang, Songtao Lu, Zengfeng Huang, and Ziping Zhao
Graph Signal Processing Workshop (GSP) 2024
14. A Joint Graph Signal and Laplacian Denoising Network
Zepeng Zhang and Ziping Zhao
Asia-Pacific Signal and Information Processing Conference (ACSSC/Asilomar) 2024
15. **Accelerating Quadratic Transform and WMMSE**
Kaiming Shen, Ziping Zhao, Yannan Chen, **Zepeng Zhang**, and Hei Victor Cheng
IEEE International Symposium on Information Theory (ISIT) 2024

16. **Enhancing the Efficiency of WMMSE and FP for Beamforming by Minorization-Maximization**
Zepeng Zhang, Ziping Zhao, and Kaiming Shen
IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) 2023
17. **Designing Graph Neural Networks via Algorithm Unrolling**
Zepeng Zhang and Ziping Zhao
International KDD Workshop on Deep Learning on Graphs: Methods and Applications 2022
18. **Weighted Sum-Rate Maximization for Multi-Hop RIS-Aided Multi-User Communications: A Minorization-Maximization Approach**
Zepeng Zhang and Ziping Zhao
IEEE International Workshop on Signal Processing Advances in Wireless Communications (SPAWC) 2021
[Best Student Paper Award Finalist](#)
19. **Vast Portfolio Selection With Submodular Norm Regularizations**
Zepeng Zhang, and Ziping Zhao
European Signal Processing Conference (EUSIPCO) 2021
20. **Multi-Period Portfolio Optimization for Financial Index Tracking**
 Xiuyuan Huang, **Zepeng Zhang**, and Ziping Zhao
Asilomar Conference on Signals, Systems, and Computers (ACSSC/Asilomar) 2021
21. **Scalable Financial Index Tracking With Graph Neural Networks**
Zepeng Zhang, and Ziping Zhao
IEEE Statistical Signal Processing Workshop (SSP) 2021
22. **A Deep Learning-Aided Approach to Portfolio Design for Financial Index Tracking**
Zepeng Zhang, and Ziping Zhao
Asilomar Conference on Signals, Systems, and Computers (ACSSC/Asilomar) 2020

PREPRINT

1. **Discerning and Enhancing the Weighted Sum-Rate Maximization Algorithms in Communications**
Zepeng Zhang, Ziping Zhao, Kaiming Shen, Daniel P Palomar, and Wei Yu
 arXiv preprint arXiv:2311.04546, 2023
2. **Towards Understanding Graph Neural Networks: An Algorithm Unrolling Perspective**
Zepeng Zhang and Ziping Zhao
 arXiv preprint arXiv:2206.04471, 2022
3. **Rate Maximizations for Reconfigurable Intelligent Surface-Aided Wireless Networks: A Unified Framework via Block Minorization-Maximization**
Zepeng Zhang and Ziping Zhao
 arXiv preprint arXiv:2105.02395, 2021

AWARDS AND HONORS

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|------|---|
| 2021 | Best Student Paper Award Finalist in IEEE International Workshop on Signal Processing Advances in Wireless Communications (SPAWC) |
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- 2019 Yu Gang Scholarship (Top 1%, Rank 1st out of the department), WHU
Outstanding Student First-Class Scholarship (Top 5%), WHU
- 2018 SAMSUNG Scholarship (Top 1%, Rank 1st out of the department), WHU
Outstanding Student First-Class Scholarship (Top 5%), WHU
1st prize of RoboCup China Open
1st prize of Asia and Pacific Mathematical Contest in Modeling
3rd prize of The Chinese Robot Skills Competitions
- 2017 Outstanding Student Second-Class Scholarship (Top 15%), WHU
1st prize of The Chinese Mathematics Competitions, Hubei Prov.

TEACHING ASSISTANT

EPFL

CIVIL 426: Machine learning for predictive maintenance applications 2023-24 Fall, 2024-25 Fall, 2025-26 Fall

CIVIL 332: Data Science for infrastructure condition monitoring 2023-24 Spring, 2024-25 Spring

MATH 106: Math Analyse II, 2024-25 Spring

ShanghaiTech

CSI82: Introduction to Machine Learning 2021-22 Fall

SI 231: Matrix Analysis and Computations 2020-21 Spring, 2021-22 Spring

ACADEMIC SERVICE

Journal Reviewer: IEEE Transactions on Wireless Communications, Mechanical Systems and Signal Processing, IEEE Open Journal of Signal Processing, Neurocomputing

Conference Reviewer: ICML, NeurIPS, LOG, AISTATS, WWW, ICASSP, ECAI, MLSP

SKILLS

Python, MATLAB, CVX, PyTorch, PyG, LaTeX

Toefl: 105 (Reading: 30; Listening: 27; Speaking: 22; Writing: 26)